

B. Pressure Exposure and Elevation Coefficient, K_z

Use Ground Surface Roughness Category C unless Ground Surface Roughness Category D applies. For noise and perimeter walls, the structure height, Z , used in calculating K_z may be taken less than 33-feet provided that K_z is greater than or equal to 0.85.

C. Gust Effect Factor, G

Delete **LRFD** Table 3.8.1.2.1-1. Delete the definition of G shown under **LRFD** Equation 3.8.1.2.1-1 and replace with the following:

G = gust effect factor. Use 0.85 for all load combinations and all structures, including noise walls and perimeter walls.

Commentary: The term all structures does not include ancillary structures (structural support for signs, signals, lighting, ITS, and tolling). See SDG 2.4.2.

D. Wind Loads on Noise and Perimeter Walls

Treat noise and perimeter walls as sound barriers in accordance with **LRFD**. Noise and perimeter walls may be designed assuming the wind pressure is applied in incremental strips (e.g. per each foot of wall height).

E. Phase Construction

SDG 2.4.1 applies to permanent portions of phase constructed bridges that support traffic in a temporary condition.

2.4.2 Wind Loads on Ancillary Structures

See **Volume 3** for wind loading on structural supports for signs, signals, lighting, ITS, and tolling.

2.4.3 Wind Loads During Construction (LRFD 3.4.2)

- A. See also **SDG 6.10** Erection Scheme and Beam/Girder Stability.
- B. Use construction wind loads to evaluate beam/girder stability during construction.
- C. Calculate wind loads during construction per **LRFD** Equation 3.8.1.2.1-1 using the load factors γ_{ws} and design wind speeds (V) in Table 2.4.3-1, the drag coefficient, (C_D), in **SDG 2.4.3.D** or **SDG 2.4.3.E**, the Pressure Exposure and Elevation Coefficient (K_z) in **SDG 2.4.1.B** and the Gust Effect Factor (G) in **SDG 2.4.1.C**.